

Sayısal Sistemler Laboratuvarı

Kombinasyonel Devreler3- H9DM1

Dr. Meriç Çetin
versiyon091025

Deney föyü

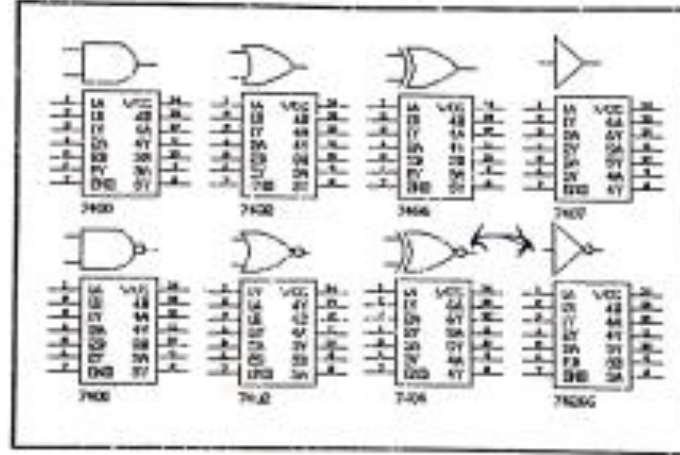


T. C

PAMUKKALE ÜNİVERSİTESİ
MÜHENDİSLİK FAKÜLTESİ

ELEKTRİK - ELEKTRONİK MÜHENDİSLİĞİ BÖLÜMÜ

LOJİK DEVRELER LABORATUVARI DENEY KILAVUZU



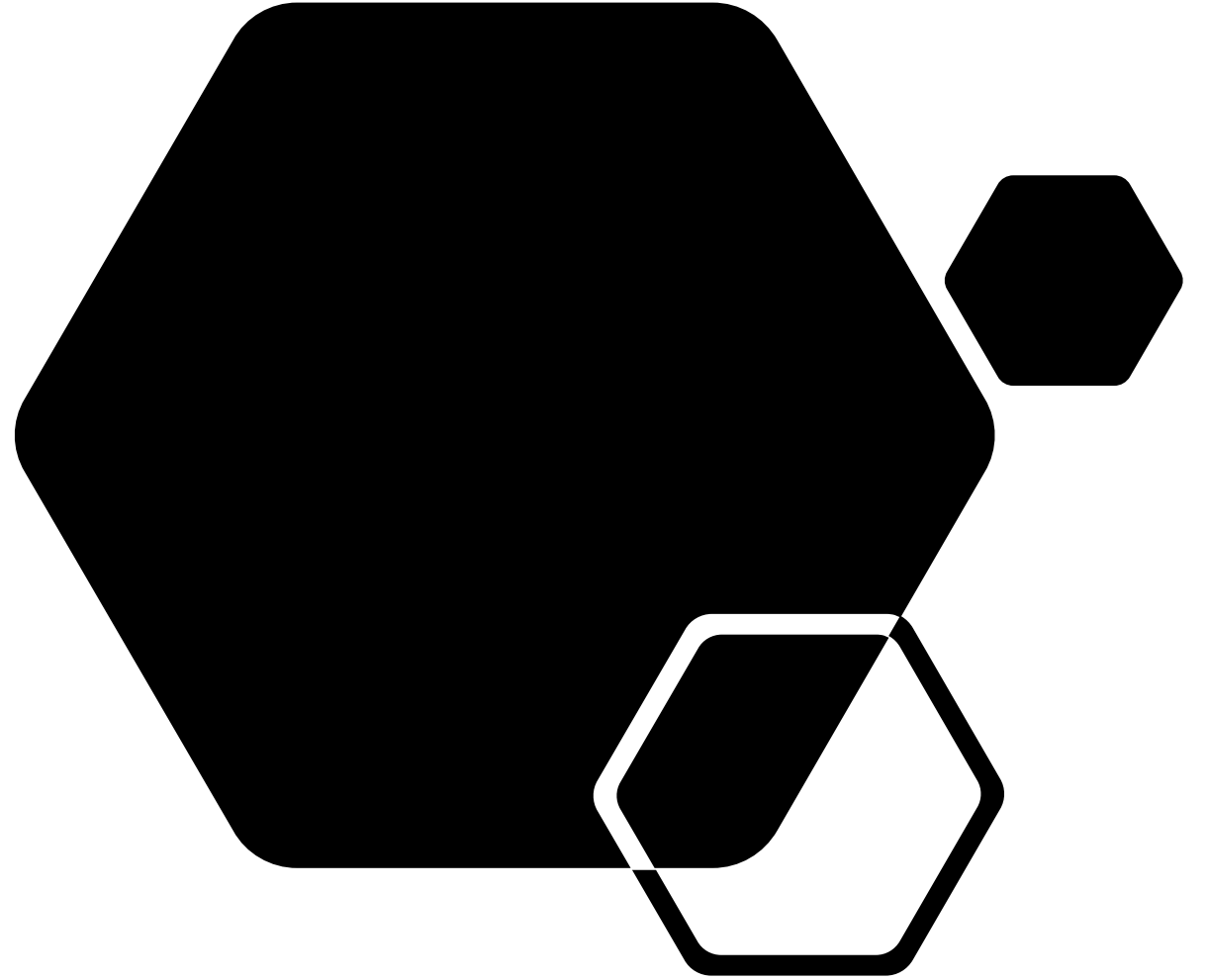
Hazırlayanlar:

Doç. Dr. Mustafa TEMİZ

Doç. Dr. Rafig SAMEDOV

Deney 12

Tam Çıkarıcı Devre



Tam Çıkarıcı Devre

b) Tam Çıkarıcı (Full Subtractor)

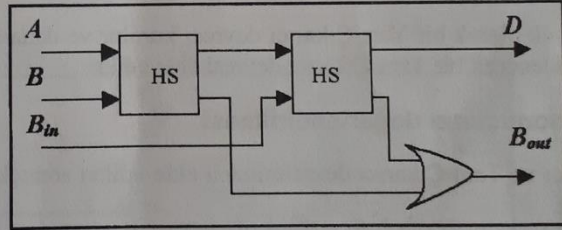
Aşağıda tasarım yoluyla bir Tam Çıkarıcının elde edilmesi görülmektedir.

1. Matematik modeli

$$D = (A \oplus B) \oplus B_{in}$$

$$B_{out} = \bar{A} \cdot B + (\bar{A} \oplus B) \cdot B_{in}$$

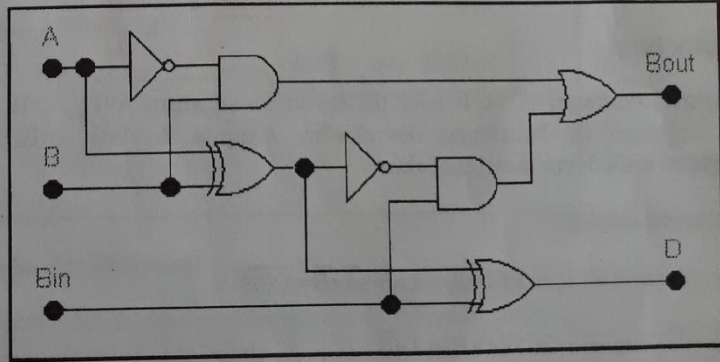
2. Fonksiyonel şeması



3. Doğruluk tablosu

INPUTS			OUTPUTS	
A	B	B _{in}	D	B _{out}
L	L	L	L	L
L	L	H	H	H
L	H	L	H	H
L	H	H	L	H
H	L	L	H	L
H	L	H	L	L
H	H	L	L	L
H	H	H	H	H

4. Prensip şeması

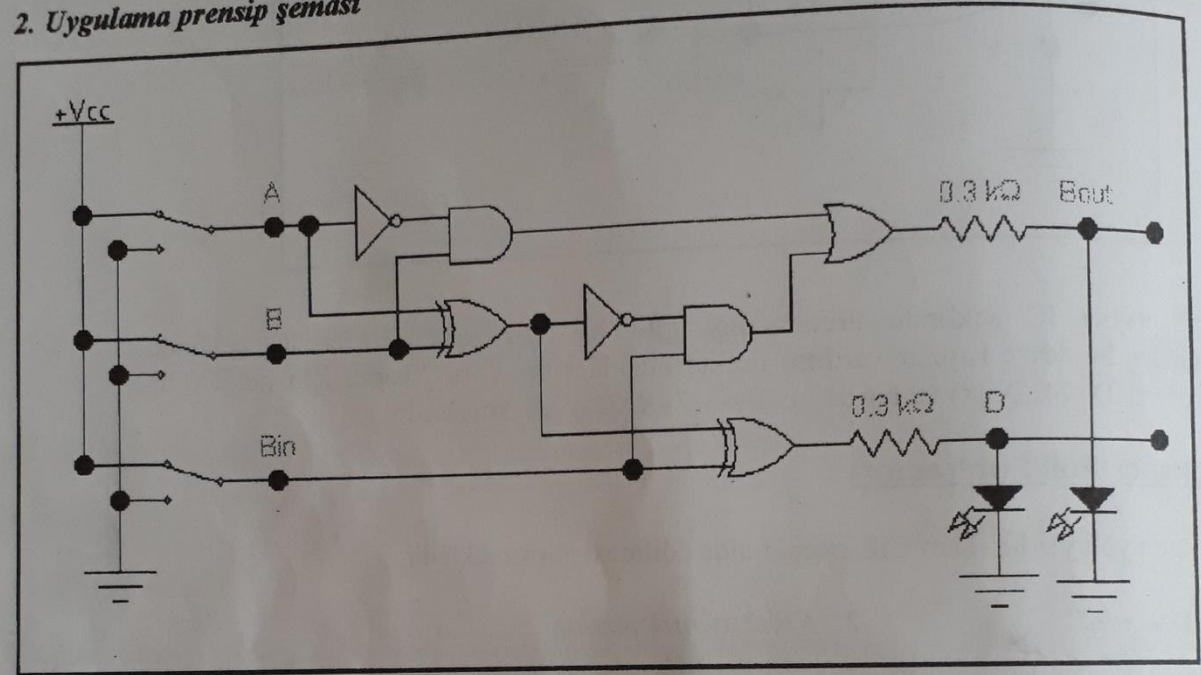


A. Deneyin Yapılışı

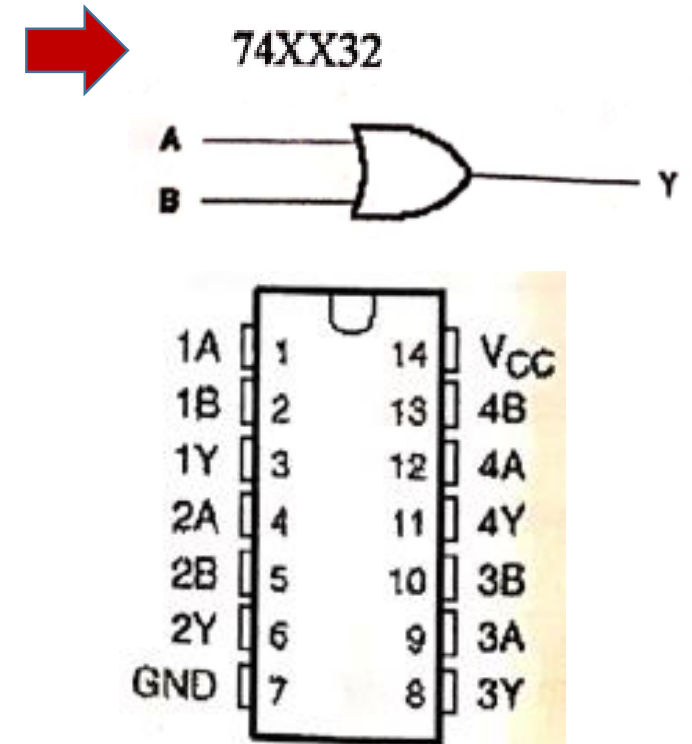
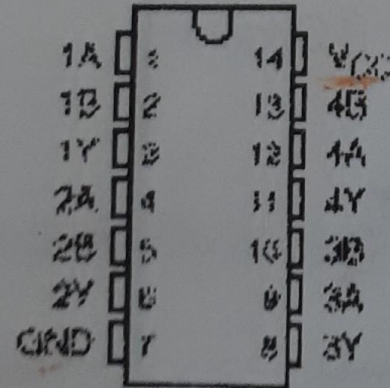
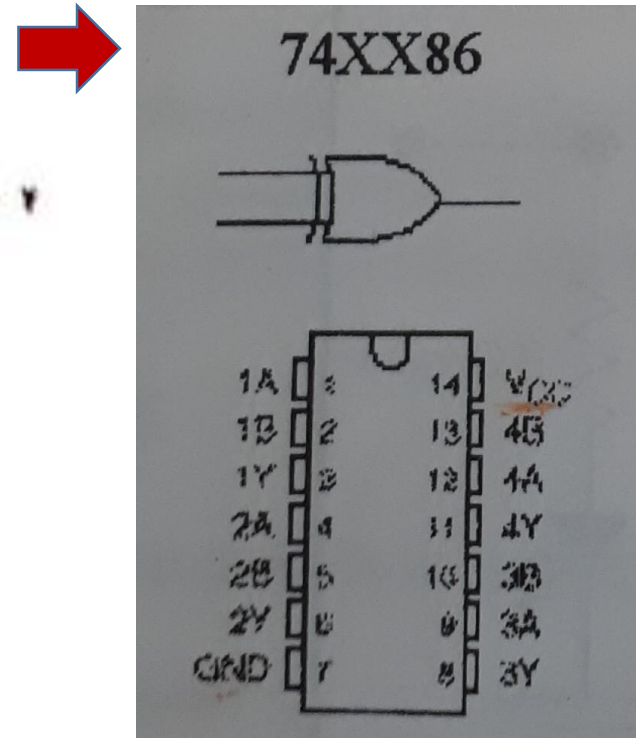
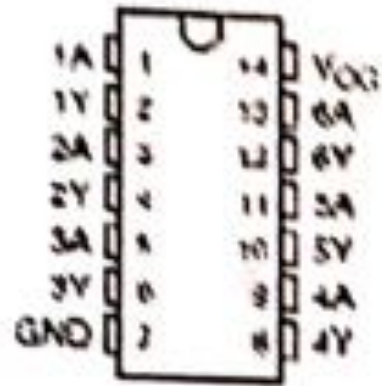
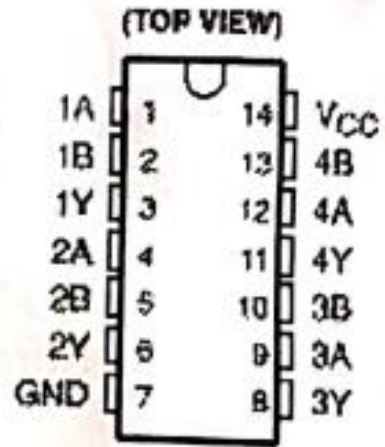
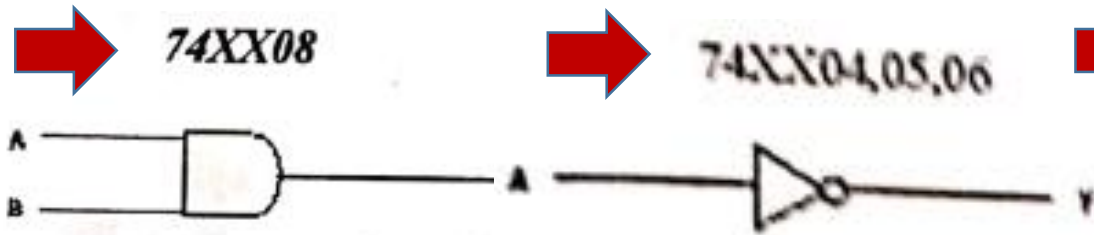
1. Deney için gereken elemanlar:

1 adet 74XX86, 1 adet 74XX08, 1 adet 74XX32, 1 adet 74XX04, 4 adet LED, 4 adet 0,3 kΩ'lık direnç, 4 adet 100 nF'lık d.k. kondansatör ve yeterli sayıda atlama teli.

2. Uygulama prensip şeması



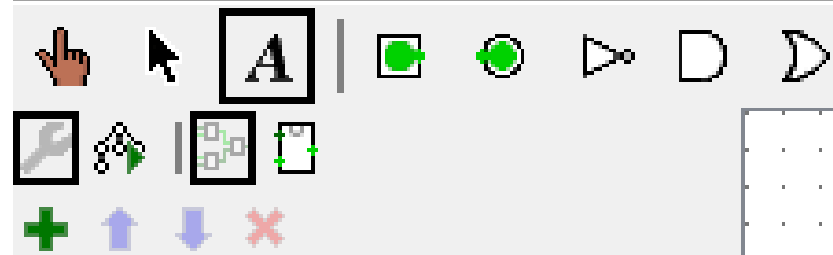
AND Kapısı NOT Kapısı EXOR Kapısı OR Kapısı



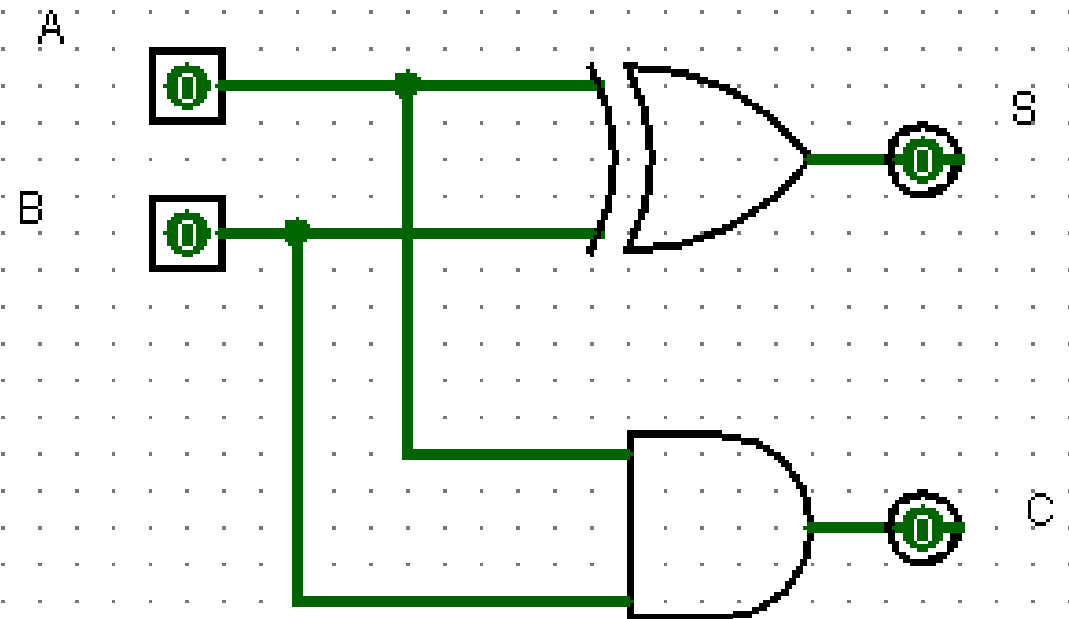
Logisim Uygulamaları

Logisim: main of Untitled

File Edit Project Simulate Window Help



- Untitled*
- main
- Wiring
- Gates
 - NOT Gate
 - Buffer
 - AND Gate
 - OR Gate
 - NAND Gate
 - NOR Gate
 - XOR Gate
 - XNOR Gate
 - Odd Parity
 - Even Parity
 - Controlled Buffer



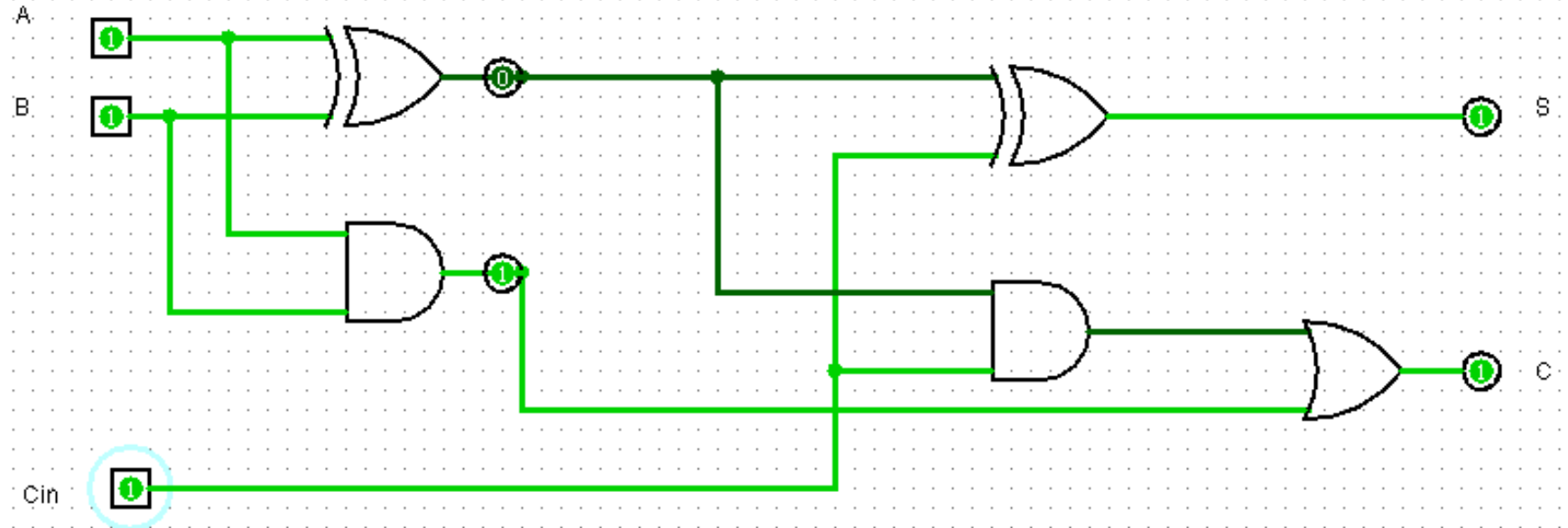
Yarı Toplayıcı Deyre

Logisim: main of Untitled

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 - Even Parity
 - Controlled Buffer
 - Controlled Inverter
 - Plexers
 - Arithmetic
 - Memory
 - Input/Output
 - Base

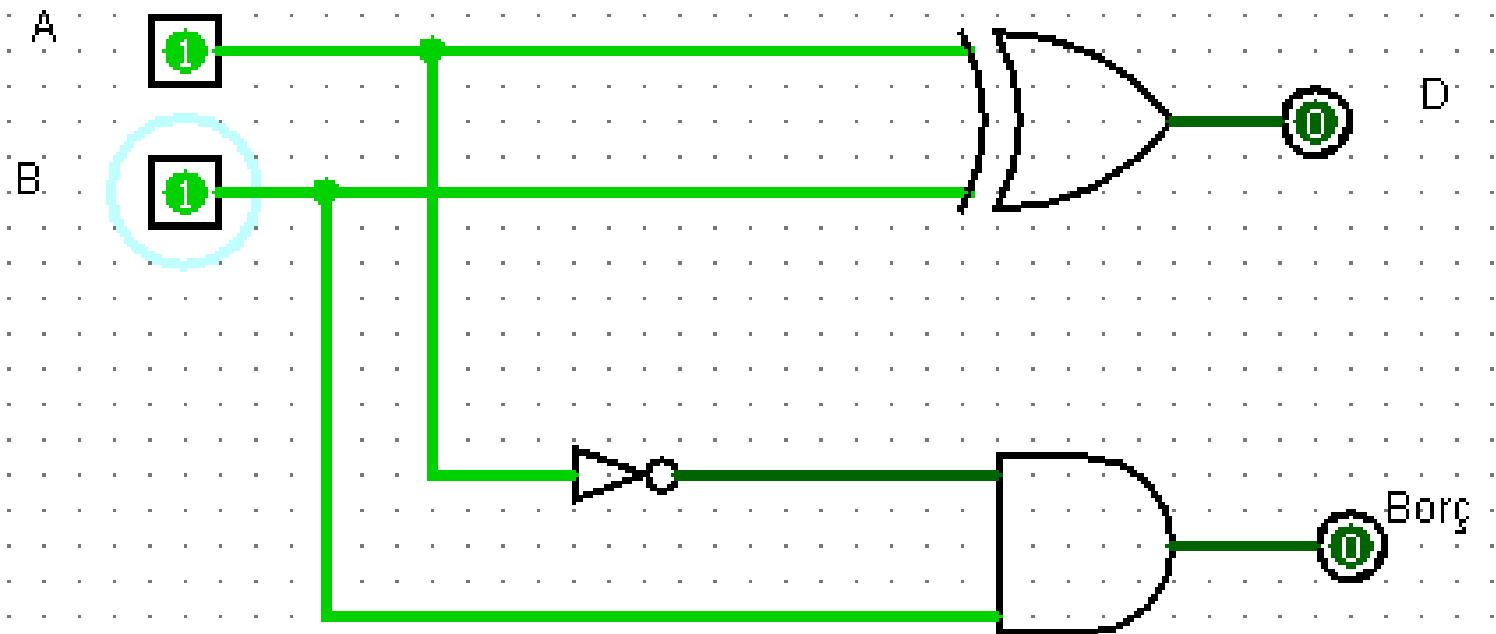


Tam Toplayıcı Devre

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- tt*
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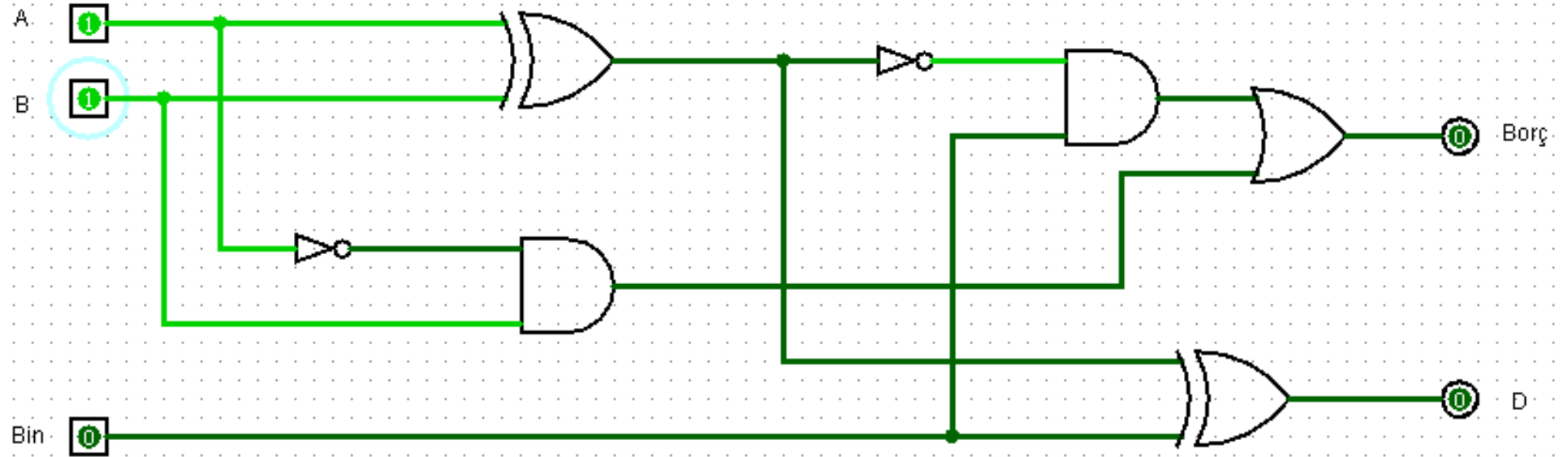


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Tam Çıkarıcı Devre

Logisim: main of tt

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